Ex:NO:01

Basic networking Commands in Linux and Windows operating system.

Aim:

To start and execute basic networking Commands in windows and linux operating system for viewing network configuration, testing connectivity, checking route information, monitoring network status and troubleshooting network related issues.

Windows:

* IP Config:-

  Display network configuration.

- Windows command - ipconfig
- Linux Command - ip addr or ifconfig

* Ping:-

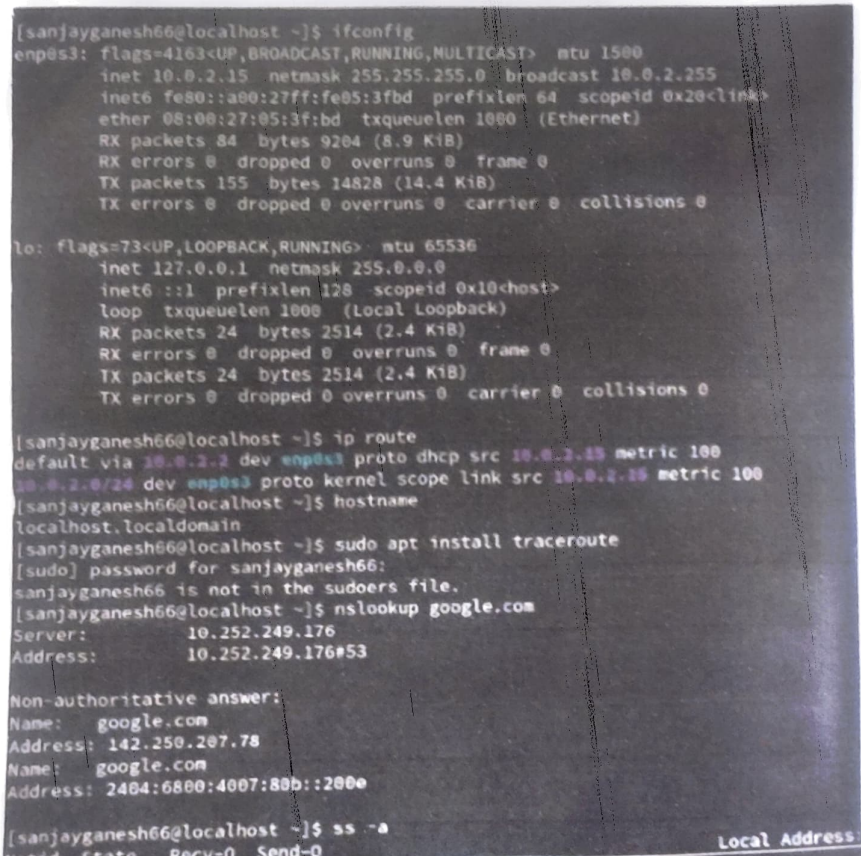
  Tests connectivity to a remote host.

- Windows command - Ping google.com
- Linux Command - Ping google.com

* Trace Route:

  Display the path packets take to reach a destination.

- Windows command - Tracert google.com
- Linux Command - Traceroute google.com



```
[sanjayganesh66@localhost ~]$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::a00:27ff:fe05:3fbd  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:05:3f:bd  txqueuelen 1000  (Ethernet)
        RX packets 84  bytes 9204 (8.9 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 155  bytes 14828 (14.4 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 24  bytes 2514 (2.4 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 24  bytes 2514 (2.4 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

[sanjayganesh66@localhost ~]$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
[sanjayganesh66@localhost ~]$ hostname
localhost.localdomain
[sanjayganesh66@localhost ~]$ sudo apt install traceroute
[sudo] password for sanjayganesh66:
sanjayganesh66 is not in the sudoers file.
[sanjayganesh66@localhost ~]$ nslookup google.com
Server:         10.252.249.176
Address:        10.252.249.176#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.207.78
Name:   google.com
Address: 2404:6800:4007:80b::200e

[sanjayganesh66@localhost ~]$ ss -a
Netid  State   Recv-Q  Send-Q                                  Local Address:
```

```
[sanjayganesh66@localhost ~]$ ss -a
Netid  State   Recv-Q  Send-Q                  Local Address:
Port                                Peer Address:Port
nl     UNCONN  0       0                                    rtnl:
NetworkManager/927                  *
nl     UNCONN  0       0                                    rtnl:
evolution-calen/2694                *
nl     UNCONN  0       0                                    rtnl:
evolution-addre/2739                *
nl     UNCONN  0       0                                    rtnl:
gnome-software/2819                 *
nl     UNCONN  0       0                                    rtnl:
kernel                              *
nl     UNCONN  0       0                                    rtnl:
xdg-desktop-por/2982                *
nl     UNCONN  0       0                                    rtnl:
avahi-daemon/761                    *
nl     UNCONN  0       0                                    rtnl:
goa-daemon/2682                     *
nl     UNCONN  0       0                                    rtnl:
gnome-software/2819                 *
nl     UNCONN  0       0                                    rtnl:
xdg-desktop-por/2982                *
nl     UNCONN  0       0                                    rtnl:
evolution-addre/2739                *
nl     UNCONN  0       0                                    rtnl:
evolution-calen/2694                *
nl     UNCONN  0       0                                    rtnl:
goa-daemon/2682                     *
nl     UNCONN  0       0                                    rtnl:
NetworkManager/927                  *
nl     UNCONN  0       0                                    tcpdiag:
avahi-daemon/761                    *
nl     UNCONN  768     0                                    tcpdiag:
                                    *
                                                             xfrm:
```

* NETSTART :

Display active network connections and listening ports.

- Windows Command - netstat / netstat - a
- Linux command - ss - a / netstat - a

* ARD:

Display the ARP Cache.

- Windows commands - arp - a
- Linux commands - arp - a / ip neigh

* Host name :

Displays the system host name.

- Windows commands - hostname
- Linux commands - host name

* Route :-

Displays the system or modified the routing table

- Windows commands - route print
- Linux command - ip route

```
[sanjayganesh66@localhost ~]$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT
 group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mo
de DEFAULT group default qlen 1000
    link/ether 08:00:27:05:3f:bd brd ff:ff:ff:ff:ff:ff
[sanjayganesh66@localhost ~]$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::a00:27ff:fe05:3fbd  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:05:3f:bd  txqueuelen 1000  (Ethernet)
        RX packets 140  bytes 13888 (13.5 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 213  bytes 19620 (19.1 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 26  bytes 2592 (2.5 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 26  bytes 2592 (2.5 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

[sanjayganesh66@localhost ~]$ cat /etc/resolv.conf
# Generated by NetworkManager
nameserver 10.252.249.176
[sanjayganesh66@localhost ~]$ ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT
 group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mo
de DEFAULT group default qlen 1000
    link/ether 08:00:27:05:3f:bd brd ff:ff:ff:ff:ff:ff
[sanjayganesh66@localhost ~]$ ss -tuln
Netid   State    Recv-Q   Send-Q     Local Address:Port     Peer Address:Port
udp     UNCONN   0        0              127.0.0.1:323           0.0.0.0:*

        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 26  bytes 2592 (2.5 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 26  bytes 2592 (2.5 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

[sanjayganesh66@localhost ~]$ cat /etc/resolv.conf
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1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT
 group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mo
de DEFAULT group default qlen 1000
    link/ether 08:00:27:05:3f:bd brd ff:ff:ff:ff:ff:ff
[sanjayganesh66@localhost ~]$ ss -tuln
Netid   State    Recv-Q   Send-Q     Local Address:Port     Peer Address:Port
udp     UNCONN   0        0              127.0.0.1:323           0.0.0.0:*
udp     UNCONN   0        0            0.0.0.0:5353            0.0.0.0:*
udp     UNCONN   0        0              [::1]:323               [::]:*
udp     UNCONN   0        0              [::]:5353               [::]:*
udp     UNCONN   0        0            127.0.0.1:631           0.0.0.0:*
tcp     LISTEN   0        4096         0.0.0.0:22              0.0.0.0:*
tcp     LISTEN   0        128              [::]:22                 [::]:*
tcp     LISTEN   0        128              [::1]:631               [::]:*
tcp     LISTEN   0        4096
[sanjayganesh66@localhost ~]$ curl google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html;charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://www.google.com/">here</A>.
</BODY></HTML>
[sanjayganesh66@localhost ~]$
```

* MAC ADDRESS:

- Windows command: getmac
- Linux command - ip link

Result:

Thus the basic networking command using windows command prompt and in Linux terminal are executed successfully.